Monetary Policy, Endogenous Bill Supply, and the Liquidity Premium

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Abstract

Liquidity premia on near-money assets (e.g. Treasury bills) co-move positively with monetary policy rates, a pattern usually read as the opportunity cost of money. I show that a cost-minimizing Treasury tilts issuance toward bills when policy eases, so part of the co-movement is issuance-mediated. A minimal model maps instrumental-variable regressions into a decomposition of the pass-through. Estimating the issuance response directly, point estimates attribute roughly one quarter of the effect of the policy rate on the premium to endogenous bill supply. The results inform the determination of US assets' liquidity premia, an increasingly central input in macro and finance.

Keywords: liquidity premium; convenience yield; monetary policy; debt management.
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1 Introduction

Liquidity premia on near-money assets such as Treasury bills rise and fall with the monetary policy rate. The standard interpretation is an opportunity-cost channel: the premium paid for liquidity services tracks the cost of holding money, which the central bank sets (Nagel, 2016; Drechsler et al., 2017). On this reading, the policy rate is the primary instrument governing near-money pricing, and asset supplies are second order outside stress periods.

This paper shows that the observed co-movement partly reflects a distinct mechanism: the Treasury's own borrowing decisions respond to monetary policy. A cost-minimizing debt manager issues more bills when policy eases and the yield curve steepens, and this endogenous supply response amplifies the co-movement between the premium and the rate. The slope of premia on rates is therefore a joint product of monetary policy and the debt-management rule, not a structural feature of money demand.

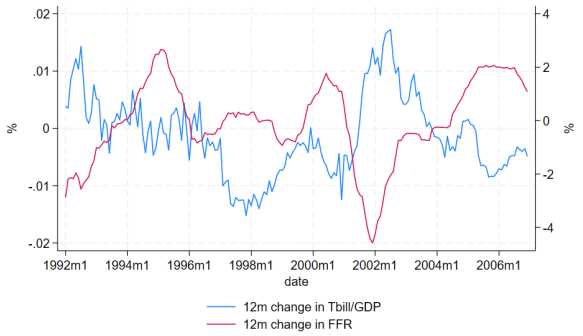
I make three contributions within a deliberately minimal framework. First, I document that bill issuance responds systematically to the monetary stance. Second, a model in which the premium arises from a CES liquidity aggregator (Nagel, 2016; Krishnamurthy and Li, 2023) and the Treasury trades off funding costs against rollover risk (Greenwood et al., 2015a) delivers the issuance response as a proposition and, crucially, an exact mapping from instrumental-variable regression coefficients to the decomposition of the total pass-through into opportunity-cost and issuance-mediated components. Third, I estimate the issuance response directly and decompose the pass-through: point estimates attribute about one quarter to endogenous supply.

The papers introducing convenience yields either abstract from monetary policy (Krishnamurthy and Vissing-Jorgensen, 2012; Greenwood et al., 2015a) or treat issuance as exogenous (Nagel, 2016; Drechsler et al., 2017). That bill supply moves premia is by now well established (Krishnamurthy and Li, 2023; d'Avernas and Vandeweyer, 2024; Phillot, 2025; Cuevas, 2025); the contribution here is that supply is itself a channel of monetary transmission, which changes how the reduced-form co-movement should be read.

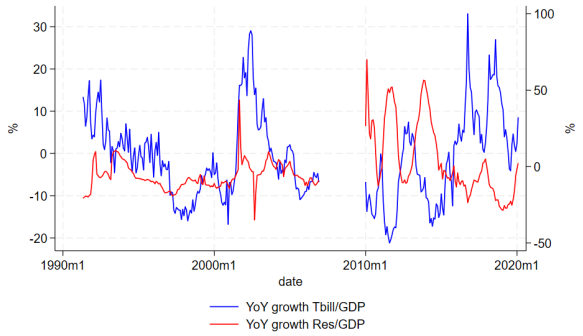
The policy implications are immediate. Liquidity premia lower short-term equilibrium interest rates (Del Negro et al., 2019), are at the center of unconventional monetary policy transmission in Krishnamurthy and Vissing-Jorgensen (2011), expand the government's fiscal capacity (Reis, 2022), and can explain exchange rate and capital flows puzzles (Kekre and Lenel, 2024). If issuance-mediated effects are material, then the policy mix matters: rate cuts that lower premia may be amplified (or confounded) by issuance tilts, calling for greater awareness between monetary operations and debt management.

Figure 1: Monetary policy and T-bill supply

(a) Federal funds rate and T-bill/GDP



(b) T-bill and reserves



Notes: variables are in 12m changes. Reserves refer to central bank reserves. Panel (a) omits the zero lower bound and the floor regime. Panel (b) omits 2007–2009 to ease visibility.

2 Two facts about issuance

Fact 1: *bill supply moves against the policy rate.* Over 1991–2008, twelve-month changes in the T-bill/GDP ratio are strongly negatively correlated with twelve-month changes in the federal funds rate (−0.56; Figure 1a): easing episodes that lower the premium see issuance tilt toward bills. The pattern is specific to the money-like margin: long-term debt shows no comparable correlation (Section 4). Thus, prices and quantities move in opposite directions—the premium falls as bill supply expands after rate cuts—implying that monetary policy shifts the *supply* of near-money, not only its demand.

Fact 2: *the issuance response adapts to the monetary regime.* In the corridor period (1991–2006), bill growth co-moves positively with reserve growth (0.46), consistent with Figure 1a. After 2008, the correlation turns sharply negative (−0.73; Figure 1b): this negative correlation is consistent with the issuance response working through quantities rather than prices. During QE and the zero lower bound, as the central bank bought long-term debt, the government minimized interest costs by tilting borrowing toward longer horizons (Greenwood et al., 2015b).[1]

[1]Result holds when using deposits instead of reserves (correlation of 0.74).

3 Framework

Pricing. The representative investor maximizes

$$\mathbb{E}_0\left[\sum_{t=1}^{\infty}\beta^t u(C_t, Q_t)\right] \tag{1}$$

where the liquidity aggregate is composed of deposits and T-bills,

$$Q_t = \left((1-\lambda_t)\left(\frac{D_t}{P_t}\right)^{\rho} + \lambda_t\left(\frac{B_t}{P_t}\right)^{\rho}\right)^{1/\rho} \tag{2}$$

where D_t are deposits, B_t are T-bills, P_t is the price level, λ_t is the relative liquidity of bills, and ρ governs substitutability; Krishnamurthy and Li (2023) document that deposits and Treasuries are imperfect substitutes ($\rho < 1$), and the CES form follows Nagel (2016).

Investors face the budget constraint

$$\begin{aligned}\underbrace{D_{t-1}(1+i^d_{t-1})}_{\text{deposit return}} + \underbrace{B_{t-1}(1+i^b_{t-1}) + B^n_{t-1}(1+i^n_{t-1})}_{\text{bond return}} + \underbrace{L_{t-1}(1+i_{t-1})}_{\text{risk-free lending return}} + \underbrace{I_t}_{\text{income}} + \\ \underbrace{\Pi_t}_{\text{bank's profits}} = \underbrace{P_t C_t}_{\text{consumption}} + \underbrace{D_t + B_t + B^n_t + L_t}_{\text{investment}}\end{aligned} \tag{3}$$

where i_t is the risk-free (illiquid) borrowing or lending rate (from banks), i^d_t the deposit rate, and i^b_t the one-period bond yield. Loans and i_t govern the intertemporal consumption decision (Euler equation).[2]

The amount of deposits that banks can create is constrained by their reserves holdings, R_t, according to $D_t \leq \phi R_t$ where $\phi > 1$ (reserve requirements or precautionary needs). Banks pay a deposit rate $i^d_t = \eta i_t$, $0 \leq \eta \leq 1 - 1/\phi$ [3]. Its balance sheet is $D_t = L_t + R_t$.

The path of R_t is chosen by the central bank to meet an interest rate operating target for i_t according to the policy rule $i^*_t = \psi(P_t/P^*_t; \xi_t)$ (with P^*_t a price-level target and ξ_t a shock) and the FOC of deposits, $u'_Q(C_t, Q_t)Q_t^{1-\rho}(1-\lambda_t)(\frac{D_t}{P_t})^{\rho-1} = u'_C(C_t, Q_t)\frac{i_t - i^d_t}{1+i_t}$. Together, these deliver that R_t and D_t fall (increase) when policy tightens (eases), resembling a corridor system of scarce reserves.

Taking the ratio of the first-order conditions for bills and deposits—so that the price level

[2]The key link between monetary policy and endogenous issuance works regardless of who demands liquidity: households (here) or banks (as in (Lenel et al., 2019; Piazzesi et al., 2021)).

[3]Competitive banks constrained by reserves ($D_t \leq \phi R_t$) deliver $\eta = 1 - 1/\phi$, and $\eta < 1 - 1/\phi$ with monopolistic banks (Nagel, 2016).

cancels—the liquidity premium $LP_t \equiv i_t - i_t^b$ satisfies

$$LP_t = i_t\,\kappa(B_t/D_t;\lambda_t), \qquad \kappa(x;\lambda) \equiv (1-\eta)\tfrac{\lambda}{1-\lambda}\,x^{\rho-1}, \tag{4}$$

with $\kappa_B < 0$ (dilution). Shocks to λ_t shift premia and induce quantity responses by both authorities; they are the identification threat motivating instruments below.

Issuance. Each period the Treasury finances F_t with bills or long debt at rate i_t^n, minimizing $i_t^b B_t + i_t^n(F_t - B_t) + \Phi(B_t)$, where $\Phi' > 0, \Phi'' > 0$ captures rollover risk from concentrating issuance at short maturities, as in Greenwood et al. (2015a). Institutionally, "regular and predictable" issuance smooths coupon sizes and leaves bill offerings as the flexible margin, so Φ is the model counterpart of that framework (Garbade, 2007; Belton et al., 2018). The long rate underreacts to the policy rate, $i_t^n = \bar{\imath}^n + \theta(i_t - \bar{\imath}) + \tau_t$ with $\theta < 1$, as the expectations hypothesis implies for any mean-reverting policy rate. Using (4), the optimum satisfies

$$\Phi'(B_t) = \underbrace{(i_t^n - i_t)}_{\text{spread saving}} + \underbrace{\rho\,i_t\,\kappa(B_t/D_t)}_{\text{marginal premium}}, \tag{5}$$

where $\rho < 1$ appears because the Treasury internalizes dilution, equating *marginal* rather than average premium revenue.

Proposition 1. *Holding D, $dB/di = \big[\rho\bar{\kappa} - (1-\theta)\big] \big/ \big[\Phi'' + \rho(1-\rho)\bar{\imath}\bar{\kappa}/\bar{B}\big]$, so bill supply falls with the policy rate if and only if $1 - \theta > \rho\bar{\kappa}$. The corridor money response ($dD/di < 0$) reinforces the negative sign.*

Proof in Appendix A.2. A rate cut steepens the effective funding spread by $(1-\theta)$ per unit, making bills relatively cheap; working against this, the premium saving $i\kappa$ shrinks with i, at rate $\rho\bar{\kappa}$. For any $\rho \leq 1$, $\rho\bar{\kappa} \leq \bar{\kappa} \approx 0.04$, so the condition can fail only if $\theta > 0.96$—far above any estimate of the sensitivity of long yields to the funds rate. Cost minimization therefore tilts issuance toward bills precisely when policy eases, matching Fact 1.

From model to regressions. Linearizing (4) with $b_t \equiv \log B_t$,

$$LP_t = \bar{\kappa}\,i_t + \gamma_0\,b_t - \gamma_0\,d_t + u_t, \qquad \gamma_0 \equiv \bar{\imath}\bar{\kappa}(\rho - 1) < 0, \tag{6}$$

with u_t collecting λ_t shocks (Appendix A.2). Deposits are concentrated out: their responses to the rate and to issuance are absorbed into the coefficients below. Four objects follow (Appendix A.3): the issuance response $\delta \equiv db/di$; the *non-issuance partial* $\beta^* \equiv \partial LP/\partial i|_b$ (opportunity-cost scaling plus the money response, controlling for b_t); the supply partial

γ (γ_0 net of deposit accommodation); and the *total* pass-through of a monetary shock, $\Pi \equiv dLP/di = \beta^* + \gamma\,\delta$.

4 Evidence

The premium is the commonly used 3-month GC repo/T-bill spread (monthly, basis points); the corridor sample is 1991m5–2008m11 for three reasons: comparability with existing papers, avoiding the zero lower bound (no variation in rates), and avoiding the abundant-reserve regime (which introduces the interest on reserves as a policy rate). I instrument the policy rate with the risk-adjusted lagged federal funds futures expectation (same as in Nagel (2016) for comparability) and bill supply with month dummies exploiting issuance seasonality (Greenwood et al., 2015a; Nagel, 2016)[4]; the instrument set, controls (VIX), and estimation sample are identical across columns (2)–(4), and I report first-stage F statistics (Sanderson–Windmeijer conditional F's in column (3)). Independent evidence that supply moves premia ($\kappa_B < 0$) includes Krishnamurthy and Li (2023), d'Avernas and Vandeweyer (2024), Phillot (2025), and, for the floor regime, Cuevas (2025).

Table 1 reports four specifications. Column (1), OLS in levels, replicates the standard result: a positive rate coefficient and an insignificant supply coefficient—the basis for reading the slope as pure opportunity cost. Equations (4)–(6) explain why this is uninformative: λ_t shocks move premia and induce quantity responses by both authorities, so OLS confounds demand and supply, and the Treasury's rate-dependent issuance makes b_t correlated with i_t.

Columns (2)–(4) are in first differences with 2SLS. Column (2) estimates the total pass-through, $\widehat{\Pi} = 12.55$. Column (3) adds instrumented supply: the rate coefficient falls to $\widehat{\beta}^* = 9.20$ and the supply partial is $\widehat{\gamma} = -105.9$, both significant. Through the lens of (6), $\gamma_0 = (\rho - 1)\overline{LP}$ per log point: column (1) taken at face value implies $\rho \approx 0.72$ (Krishnamurthy and Li, 2023), while the instrumented estimate implies steeper monthly bill demand (d'Avernas and Vandeweyer, 2024)—the same simultaneity that attenuates the OLS supply coefficient overstates substitutability. Column (4) is the key new step: regressing bill-supply growth on the instrumented rate yields the issuance response $\widehat{\delta} = -0.0316$ log points (≈ 3.2 percent) per percentage point—a directly estimated, causally interpretable analogue of Fact 1—significant under HAC and identification-robust Anderson–Rubin inference. The memo rows verify the adding-up $\widehat{\Pi} = \widehat{\beta}^* + \widehat{\gamma}\widehat{\delta}$ and report the issuance-mediated share, $s = 0.27$, about one quarter of the total in point estimate (a complete set of 95% block-bootstrap CIs is reported in

[4]Year-end window-dressing demand is the standard objection; results stand when moving the year-end-dummy to the control set as robustness.

Table 1: Monetary policy, bill supply, and the liquidity premium

	(1)	(2)	(3)	(4)
Dep. var.	LP_t	ΔLP_t	ΔLP_t	Δb_t
Estimator	OLS	2SLS	2SLS	2SLS
Policy rate	6.706***			
	(1.067)			
ΔPolicy rate		12.55***	9.20***	−0.0316**
		(2.759)	(2.32)	(0.015)
log(Tbill/GDP)	−7.298			
	(9.756)			
Δ log(Tbill/GDP)			−105.9***	
			(24.1)	
Estimand	—	Π	β^*, γ	δ
First-stage F	—	38.2	44.8, 14.9	38.2
Observations	212	211	211	211
Issuance-mediated component $\widehat{\gamma}\widehat{\delta} = 3.35$; share $s = 1 - \widehat{\beta}^*/\widehat{\Pi} = 0.27$ (95% Fieller CI $[0.00, 0.58]$)				

Monthly, 1991m5–2008m11; Δ ≡ 1-month change. Instruments: ff futures surprises, month dummies; VIX control. Newey–West SEs (12 lags). First-stage F is Sanderson–Windmeijer in column (3).
Anderson–Rubin 95% set for δ: $[-0.056, -0.001]$.

Appendix A.5).

Finally, we characterize the mechanism behind $\widehat{\delta}$. Policy-rate changes move the relative cost of bill finance: a 100 bp cut widens the long-term–bill financing spread by $55 - -68$ bp. This changes the Treasury's announced maturity composition: IV-local projections show the response cumulating to -0.365 log points by month 11, the pace at which announced auction sizes adjust. Lastly, the composition moves the outstanding stock and thus the premium (Table 1, Column 3). Appendix A.6 documents each result and rules out competing mechanisms suggested by issuance blind to relative costs at monthly frequency.

5 Implications

In point estimates, endogenous issuance *amplifies* the grip of monetary policy on convenience yields ($\Pi > \beta^*$), but through a fiscal rule the central bank does not control. The premium–rate slope is therefore not invariant to debt management: as issuance becomes rigid ($\Phi'' \to \infty$, e.g. "regular and predictable" bill-share bands), $\delta \to 0$ and $\Pi \to \beta^*$. Debt management is, in effect, a second monetary instrument for near-money pricing; recognizing it changes both the measurement and the design of policy toward convenience yields.

Corollary 1. *If issuance is constrained ($B_t = \bar{B}$, e.g. a debt-limit impasse), then $\delta \equiv$*

$db/di = 0$ and the decomposition collapses to $\Pi = \beta^*$: pass-through falls to the non-issuance partial, which is a property of the pricing block alone and hence invariant to the Treasury's rule, while shifts in $\bar{B}$ move the premium through γ alone.

Proof in Appendix A.2. Constrained periods thus feature weaker policy transmission to premia and scarcity-driven premium spikes, consistent with bill-shortage episodes (d'Avernas and Vandeweyer, 2024). Conversely, if long yields co-moved (nearly) one-for-one with policy—Proposition 1's condition failing, contrary to the financing-spread response in Appendix A.6—δ would weaken or reverse and pass-through fall toward, or below, β^*: policy *less* potent than the opportunity-cost view implies (Appendix A.2).

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